

INNOVATION and DESIGN THINKING

21IDT19

MODULE-1

PROCESS OF DESIGN

Understanding Design thinking

Shared model in team-based design – Theory and practice in Design thinking – Explore presentation signers across globe – MVP or Prototyping

PROCESS OF DESIGN

What is Design Thinking?

Design thinking is a process for creative problem solving.

Design thinking has a human-centered core. It encourages organizations to focus on the people they're creating for, which leads to better products, services, and internal processes. When you sit down to create a solution for a business need, the first question should always be what's the human need behind it?

In employing design thinking, you're pulling together what's desirable from a human point of view with what is technologically feasible and economically viable. It also allows those who aren't trained as designers to use creative tools to address a vast range of challenges. The process starts with taking action and understanding the right questions. It's about embracing simple mindset shifts and tackling problems from a new direction.

Why is design thinking important?

Design thinking enables organizations to create lasting value for consumers. The process is useful in any complex system (not just design systems) because it:

1. Aims to solve a concrete human need

Using an observational, human-centric approach, teams can uncover pain points from the consumer that they hadn't previously thought of, ones that the consumer may not even be aware of. Design thinking can provide solutions to those pain points once they're identified.

2. Tackles problems that are ambiguous or difficult to define

Consumers often don't know what problem they have that needs solving or they can't verbalize it. But upon careful observation, one can identify problems based on what they see from real consumer behavior rather than simply working off of their ideas of the consumer. This helps define ambiguous problems and in turn makes it easier to surface solutions.

3. Leads to more innovative solutions

Humans are not capable of imagining things that are not believed to be possible, which makes it impossible for them to ask for things that do not yet exist. Design thinking can help surface some of these unknown pain points that would otherwise have never been known. Using an iterative approach to tackle those problems often leads to non-obvious, innovative solutions.

4. Makes organizations run faster and more efficiently

Rather than researching a problem for a long time without devising an outcome, design thinking favors creating prototypes and then testing to see how effective they are.

THE FIVE STAGES OF THE DESIGN-THINKING PROCESS

Design thinking follows a five-stage framework.

1. Empathize

In this first stage, the designer observes consumers to gain a deeper understanding of how they interact with or are affected by a product or issue. The observations must happen with empathy, which means withholding judgment and not imparting preconceived notions of what the consumer needs. Observing with empathy is powerful because it can uncover issues the consumer didn't even know they had or that they could not themselves verbalize. From this point, it's easier to understand the human need for which you are designing.

2. Define

In this second stage, you gather your observations from the first stage to define the problem you're trying to solve. Think about the difficulties your consumers are brushing up

against, what they repeatedly struggle with, and what you've gleaned from how they're affected by the issue. Once you synthesize your findings, you are able to define the problem they face.

3. Ideate

The next step is to brainstorm ideas about how to solve the problem you've identified. These ideation sessions could be in a group, where your team gathers in an office space that encourages creativity and collaboration, an innovation lab, or can be done solo. The important part is to generate a bunch of different ideas. At the end of this process, you'll come up with a few ideas with which to move forward.

4. Prototype

This is the stage that turns ideas into an actual solution. Prototypes are not meant to be perfect. The point of a prototype is to come out quickly with a concrete version of the idea to see how it is accepted by consumers. Examples of prototypes include a landing page to test consumer desire for a product or a video that demonstrates streamlined logistic processes.

5. Test

Once you give a prototyped solution to consumers, you must observe how they interact with it. This testing stage is the one in which you collect feedback on your work. The design-thinking process is an iterative, rather than linear, one. At the end of the fifth stage, you'll likely have to go back to one or several of the other stages. Perhaps the testing has shown you need to develop another prototype, for which you'd return to the fourth stage. Or perhaps it's shown that you've misdefined the consumer's needs. If so, you would have to return to an earlier stage of the process.

What industries and roles can benefit from design thinking?

While design thinking originated with designers, it is now widely used by people from all disciplines. Even among design agencies the work is famously cross-functional: IDEO and similar agencies hire non-designers—chefs, engineers, social scientists, biologists—and integrate them into their project teams to add perspective.

Our growth innovation team at WeWork comprises a designer, who focuses on applying this method for the end consumer of a project; a technologist, who uses this technique to deliver value to engineers; and a business strategist, who applies this method to deliver value for business owners and various stakeholders.

Design thinking has been used at Kaiser Permanente to overhaul the system of shift changes among nursing staff. It has helped the Singapore government make the process for securing a work pass in the nation-state easier and more human. Design thinking has been used to solve business problems at companies like Toyota, Intuit, SAP, and IBM.

One reason for the proliferation of design thinking in industries is that it's useful to break down problems in any complex system, be it business, government, or social organizations. It can be used to explore big questions about how to respond to the growth of technology and globalization, how to pivot in response to rapid change, and how to support individuals while catering to larger organizations.

Design thinking can be used by all departments in a business. It can be fostered by bright, airy physical workspaces that cater to the way employees prefer to work. To employ design thinking in all projects, managers should first define the consumers they're trying to help and then employ the five stages of design thinking to define and tackle the identified problems. Employing a design-thinking process makes it more likely a business will be innovative, creative, and ultimately more human.

UNDERSTANDING DESIGN THINKING

SHARED MODEL IN TEAM-BASED DESIGN

If we apply the analogy of Design Thinking as a journey, having good travel partners is important to safely and successfully arriving at your desired destination. The process of Design Thinking takes you into unknown and sometimes uncomfortable territory. Members of a Design Thinking team need to be open minded, curious, collaborative and allow their assumptions to be challenged, ready for change, and be adaptable. Yes, that's a lot, but it's worth the effort as it creates a great team spirit, work ethic – and ditto the end product.

Who to Take on the Ride

Team sizes may vary depending upon the challenge at hand, and may even comprise multiple teams working in tandem to tackle various parts of a more complex problem, or people or teams working alone sometimes and merging together at other times.

Cross-disciplinary teams will provide you with the best results. Teams may consist of people unfamiliar with each other, with external members brought on board either as specialists

or facilitators depending on the availability of skills. Navigating the people dynamics will most likely be one of the toughest parts of a Design Thinking process, although it is also the most rewarding as it involves the coming together of many different minds and personalities, whilst adding richness to the experience.

Build the Right Kind of Team

The right kinds of specializations are important, but specialization is not the only quality required. To make a Design Thinking project successful, we need T-shaped people. T-shaped people have a depth of knowledge and experience in their own fields but they can also reach out and connect with others horizontally and create meaningful collaborations.

A Design Thinking team should ideally be a cross/multi-disciplinary team consisting of a mix of specializations, including specialists associated with problem areas contributing but not dominating the journey. While specialists may have vast knowledge on a technical level, they are working towards solutions targeted towards non-specialists in many cases and require outside perspectives in addition to what they already know.

Taking a cross section of an organization, one will soon realize that individuals within specific departments with specialist skills in specific business functions tend to approach problem solving on their own level of experience, using the skill set they feel most comfortable with. Sales, marketing, IT, product development, customer service and HR departments all view challenges they face through the lens of their departmental functions. Bringing together teams that provide a wider view of things is important, but even more so is to encourage them to look outside their own spheres of influence to allow more holistically framed problems to be uncovered.

All too often, solutions are conceptualized by account managers and merely handed down to creative teams for execution and decoration. Often, this results in a team tackling the wrong problems. Individuals who are able to see the bigger picture in a situation are considered valuable assets, as are those who are able to connect with real people outside the organizational structures in order to understand the challenge from a market perspective.

Build The Right Team Culture

Choosing the perfect team may not always be possible, especially in situations where there is a limited pool of candidates. Within startup environments everyone needs to be involved in everything, so it may be more of a case of developing the appropriate team culture, or even project culture, in order to move forward with the Design Thinking process. Encouraging

empathy, experimentation, curiosity, courage, open-mindedness, holistic thinking, stripping away biases, and favoring ambiguity over rational clarity, are great places to start.

It is important to set some rules and encourage individuals to think big-picture thoughts before getting stuck on the executional nitty gritty. All team members should be encouraged to respect each other's inputs, in order to dig deeper and to build upon each other's findings, as opposed to trying to dominate and score with the winning idea as a solo effort. The following attributes or characteristics of Design Thinking will help you to choose and select team members for a given Design Thinking work project. Ask yourself questions such as: Which team members already possess these characteristics? Who would be open to them? Which team members possess the core expert skills required? You should, of course, practice these qualities yourself in your own leading style, but you should also print and hand out this illustration to each of the team members and explain the attributes featured therein. This will help to develop a great working culture.

ATTRIBUTES OF DESIGN THINKERS

Attribute	Description	Comment
Ambiguity	Being comfortable when things are unclear or when you don't know the answer	Design Thinking addresses wicked = ill-defined and tricky problems.
Collaborative	Working together across disciplines	People design in interdisciplinary teams.
Constructive	Creating new ideas based on old ideas, which can also be the most successful ideas	Design Thinking is a solution-based approach that looks for an improved future result.

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Curiosity	Being interested in things you don't understand or perceiving things with fresh eyes	Considerable time and effort is spent on clarifying the requirements. A large part of the problem solving activity, then, consists of problem definition and problem shaping.
Empathy	Seeing and understanding things from your customers' point of view	The focus is on user needs (problem context).
Holistic	Looking at the bigger context for the customer	Design Thinking attempts to meet user needs and also drive business success.
Iterative	A cyclical process where improvements are made to a solution or idea regardless of the phase	The Design Thinking process is typically non-sequential and may include feedback loops and cycles (see below).
Nonjudgmental	Creating ideas with no judgment toward the idea creator or the idea	Particularly in the brainstorming phase, there are no early judgments.
Open mindset	Embracing design thinking as an approach for any problem regardless of industry or scope	The method encourages "outside the box thinking" ("wild ideas"); it defies the obvious and embraces a more experimental approach.

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Level the Playing Field

While you may already have experts in certain fields or specialists working alongside generalists or less experienced individuals, something to stipulate clearly is the flat hierarchy, which is needed in Design Thinking projects. The more senior team players' knowledge and experience will definitely be important and will be necessary in order to steer the ship in the right direction so the team avoids becoming completely lost. However, this should not prevent less experienced individuals from sharing their ideas, putting forward alternative ways of seeing things, or flagging something that everyone else may have missed. The whole purpose of a multi-faceted team is to encourage wider perspectives. No one within the team should purposely be allowed to narrow things to suit his or her specific view. This illustrates how design-led innovation differs significantly from processes used in traditional business or corporate settings. For instance, just a mere insinuation a senior player may make towards a less experienced team member could result in a closed mindset and prevent ideas from flowing.

The Importance of a Good Facilitator

Having a leader to spearhead the mission is important, as there is always a tendency for team members to resort to familiar patterns, which may not suit the needs of Design Thinking. Hierarchies may set in quickly and management styles may subvert the mission. A leader who is experienced in maintaining the right mix of mindsets is essential. An ideal leader is someone who can maintain high levels of energy and enthusiasm, someone who can steer the group around obstacles, and someone who has at least some grounding in running Design Thinking or similar projects.

A Design Thinking Team Manifesto

Venturing into the unknown can make some team members feel uncomfortable, especially the business executive who has been trained to base decisions and actions on what has worked in the past. Basing decisions, strategies, and even attempts at innovation on past data means churning out much of the same, possibly with just a variation in colour or flavour.

If it's clear that the process won't be respected upfront by some of your team members, more orientation may be required or buy-in sought before getting started. In order to ensure that there is enough divergence in thinking, it is important to set some ground rules right from the start and display these on the wall. You can find inspiration in the Attributes of Design Thinking illustration above. You can actually easily turn this into your Design Thinking Team Manifesto. However, we strongly encourage you to develop your own manifesto with your team members in order to involve and engage all members in the Design Thinking process from the beginning. It

is important to allow and encourage anyone on the team to call up rules. Do so without malice and in the good positive spirit of exploration and experimentation.

The design process leader should keep the ship steering towards its goals with the help from a Design Thinking Team Manifesto. Having a guide who is not a person who can be blamed may be a good alternative for teams who are not quite ready to be led into the unknown by someone they don't yet know and trust.

The Effects of Design Thinking on Teams

Design Thinking is structured in such a way so as to set the scene for specific qualities to be unleashed within the team and to overcome obstacles that innovation teams typically face.

Some of the aims of Design Thinking's approach are to create:

- greater inclusiveness
- better team cohesion
- higher levels of collaboration and interaction
- increased creative confidence

Everyone thinks, feels, and experiences things differently. Differences are what we need. Knowing that certain activities are not natural for everyone means that some preparation, explanation, and trust-building, before the launch, can go a long way towards preventing people from jumping ship.

Some of the obstacles the process aims to overcome:

- Fear of failure
- Intimidation
- Low self esteem
- Hierarchy
- Creative blocks
- Blind spots
- Specialization bias

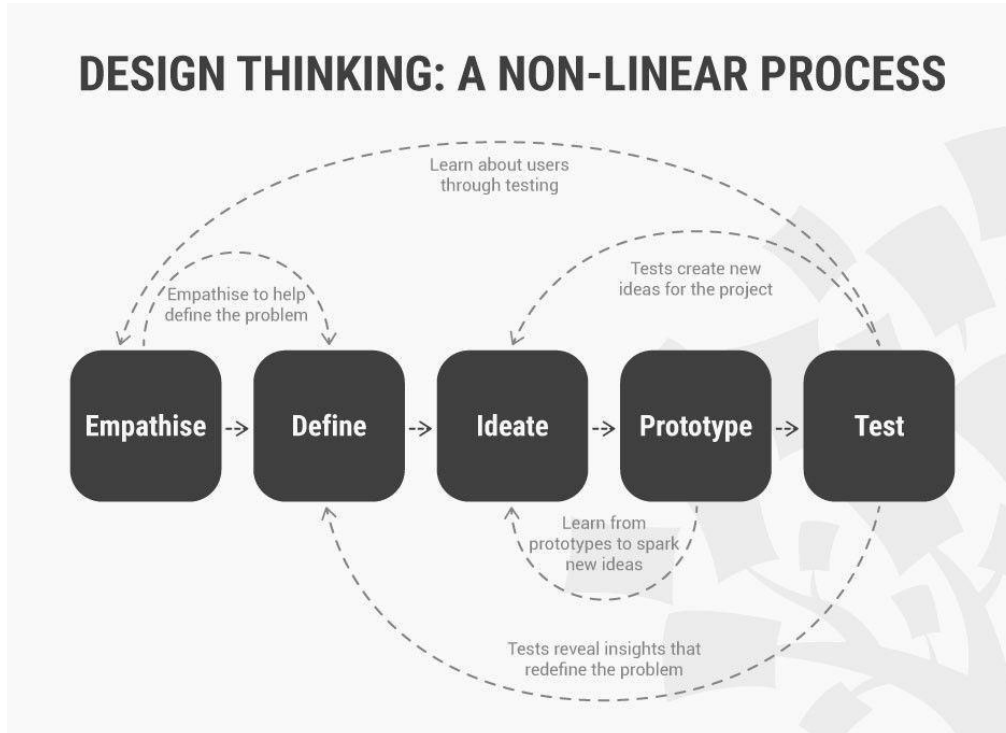
In most cases, newcomers to Design Thinking will feel intimidated, confused, and disoriented by its seemingly chaotic approach. In order to break through this negative tension, a few short exercises can be implemented to loosen things up and get people feeling positive and excited about the project.

Facilitation Step 1: Tell your Team Members about the Overall Process of Design Thinking

- Explain the five stages in the Design Thinking Process.
- Explain that the five stages of Design Thinking are not sequential steps, but different modes you can put yourself in, to iterate on your problem definition, ideas, or prototype, or to learn more about your users at any point in the project.
- Explain how Design Thinking builds a third way – combining the analytical and information-driven approach of science with the holistic, empathic and creative ways of thinking in ethnography and design.
- Explain that there are lots of proven methods that you'll apply along the way, like user interviews, brainstorm sessions, affinity clustering, sketching, user testing, etc.
- Knowing the background and underlying structure will help your team members to feel safer as they know that there's a solid background, and that chaos is not the goal even though it may feel chaotic from time to time during the process. You could remind your team members that innovation demands courage and an open mind while progressing towards the goal:

“One must still have chaos in oneself to be able to give birth to a dancing star.

– Friedrich Nietzsche



Facilitation Step 2: Break the Ice

Icebreakers are important for breaking through barriers when teams need to start working on something new, or when people don't know or trust each other well enough. Each setting is unique and the ice that needs to be broken will be different for each situation. It may be as simple as introducing people properly, or explaining the process they are about to engage in, though at times it may require more intervention.

A few icebreaker tips

- Consider barriers that may exist in the team and design your icebreakers based on these.
- Ensure that the icebreakers don't alienate anyone or create any further obstacles.
- The idea is not to embarrass but to build confidence and make people comfortable going forward.
- Icebreakers are not meant to produce winners and losers but are meant to level the playing field.
- Choose activities which suit the people and setting.

Facilitation Step 3: Get People into the Creative Mood

Co-create a story

A great way to get your Design Team members to relax and enjoy co-creation is the act of group story telling. You may begin with a completely unrelated story, maybe even something familiar to the team or something related to the challenge space being explored. Start the story off with a brief intro and move it around the team in a circle, with team members adding on their sentences as it passes on.

A set of story Cubes may help even further, especially with those who may be too shy to suggest something different. The story cubes come with a range of objects, actions, and themes that can be mixed and matched to create or encourage randomised story telling. It's also an excellent tool for getting people in the mood for brainstorming.

Visual telephone

Another similar activity which encourages visualisation rather than spoken stories is the visual telephone. The team starts off with the first player drawing a simple picture in secret, and then passing it on to the person beside her/him, who will then look at the image without showing the others, and will write down her/his interpretation of what the image relates to. The next person will then take that explanation and draw an image to show it, and so on, until the telephone has moved through the entire team. The entire team will then review the end results to find out how each person interpreted the message they received.

- Mindtools provides a good icebreaker outline with some examples for icebreakers in different settings

- Gamestorming is a book and method by Dave Gray, Sunni Brown and James Macanufo, which contains a wealth of ideas for encouraging visual thinking and action-oriented problem solving. It contains tons of advice on how to get people out of their seats and comfort zones.
- The icebreakers above are merely ways of getting people ready, but it definitely doesn't end there.

The Takeaway

- Bring together a diverse team with different thinking styles and specialisations.
- Develop an innovative team culture, which embraces inclusiveness, collaboration, and co-creation.
- Level the playing field to allow for a diverse set of perspectives to influence the process.
- Ensure the right person is in charge.
- Break the ice with some creative exercises to loosen things up.
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THEORY AND PRACTICE IN DESIGN THINKING

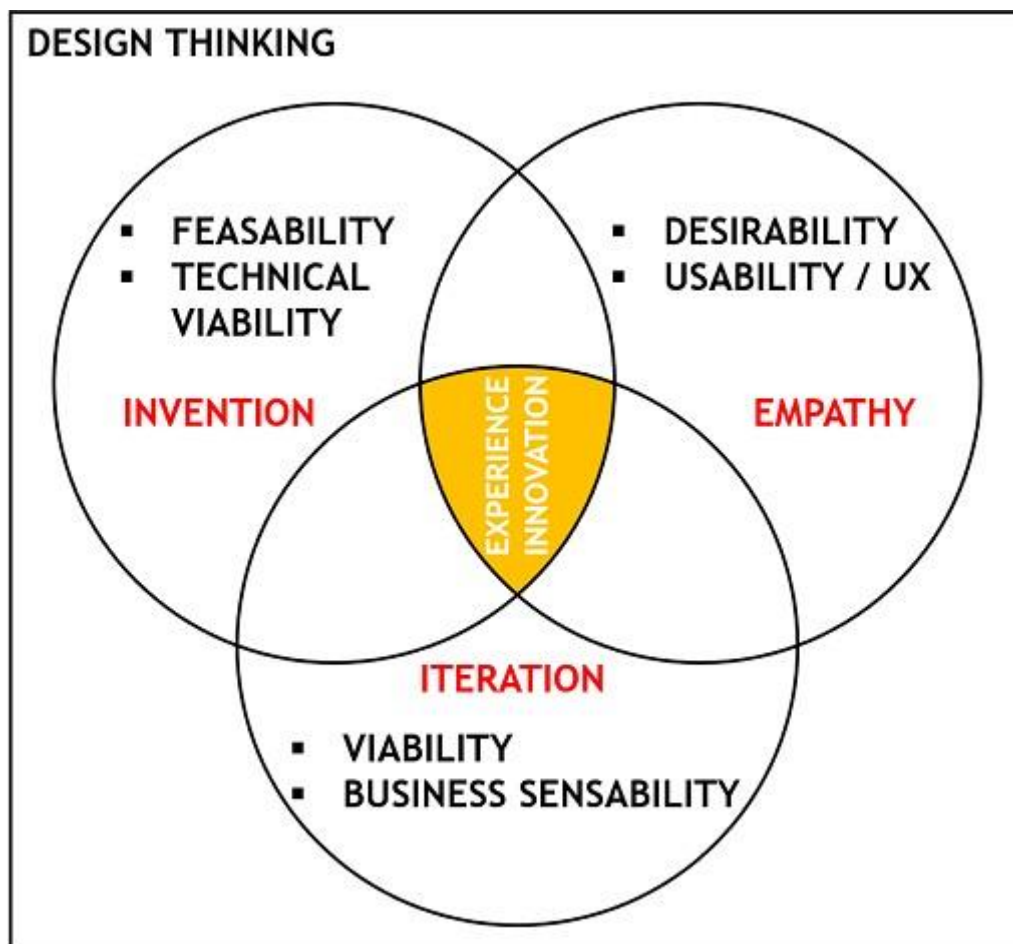
Design thinking is a human-centric methodology that requires a multidisciplinary team and can be applied by regular, non-creative' individuals

Design thinking can help develop new offers, increase levels of organizational innovation, and accelerate growth in existing businesses or new ventures (i.e. startups). Design thinking is considered a powerful approach in helping people envision new opportunities and become comfortable with uncertainty. Design thinking is a human-centric methodology that requires a multidisciplinary team and can be applied by regular, 'non-creative' individuals. 'Creative' people are not geniuses, but rather people who boost their creativity by triggering positive emotions through the activation of intrinsic motivation. Therefore, design thinking provides processes to experiment with and learn how to uncover creative insights and innovative solutions following clear and guided processes, consequentially transforming regular people into creative ones.

Design thinking is an approach for innovation, a systematic approach that aims to identify customer dis-satisfactions and build innovative solutions through iteration. Therefore, design thinking is a problem-solving approach.

Leaders now look to innovation as a principal source of differentiation and competitive advantage; they would do well...

Even though design thinking is about launching an innovation in the future — counter-intuitively — it does not start with thinking of the future, but the present. Because the objective is to ensure the desirability of the solution by addressing a real customer need. Once desirability is validated, then business viability and technical feasibility will be addressed. It is, therefore, a multidisciplinary process that favors collaboration between designers, social scientists, business people, and engineers.



Design thinking and the creation of innovative value by Tim Brown CEO of IDEO

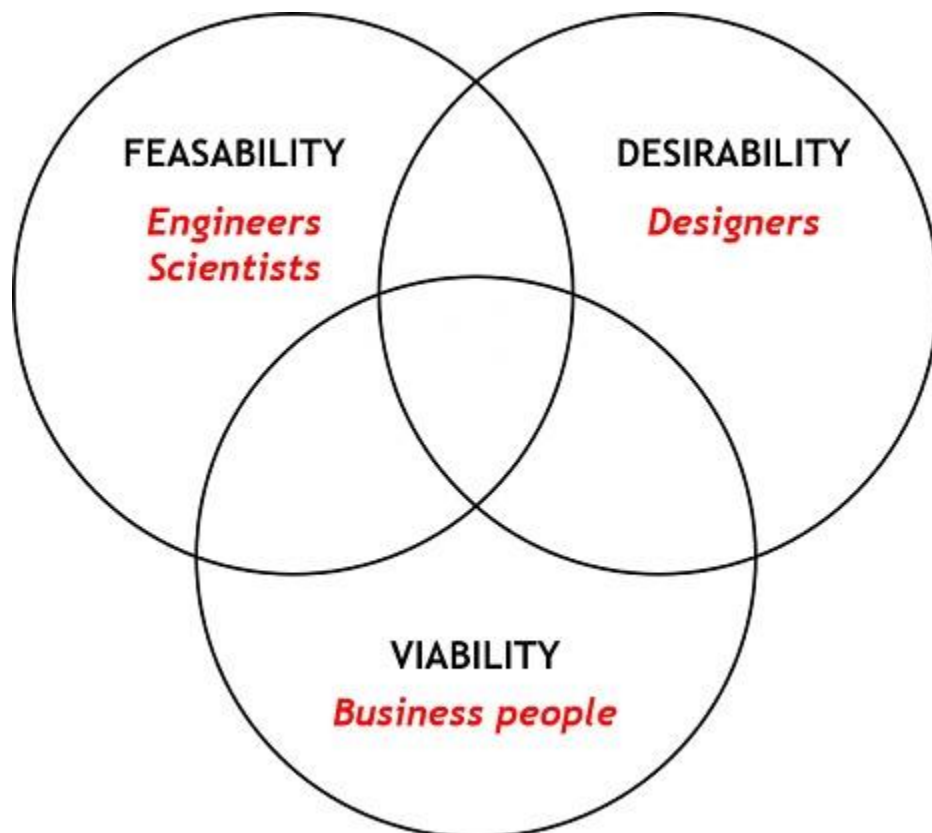
Design thinking is a systematic approach for innovation, a creativity method, a problem-forming and solving approach, and a mindset, that is based on the following three principles:

- Empathy with the users for whom the innovation is developed, to fully understand their pains and problems.
- The invention of something new that does not already exist or that enhances current solutions. It's about imagining and creating the future, rather than accepting current norms.
- Iteration, because it is hard to develop a deep understanding of user needs, requiring to test and position the imagined solution for users.

The design in design thinking

A good design is associated with a product or service that is easy to use and interact with. For example, consider your smartphones, they have been designed to be addictive, to have consumers hooked. Or consider push notifications on social media applications such as Instagram or Facebook; these notifications are triggers of interaction with the content of interest to increase engagement rates with the service. This is the design in design thinking.

The 3 elements that make smartphones so hard to put down, explained by Google's former design ethicist. For a successful design to happen, designers, scientists/engineers, and business people come together to collaborate and share skill sets in the following ways:



Design thinking and the creation of innovative value by Tim Brown CEO of IDEO (simplified version)

- Designers — ensure the desirability of the product/service, because they can empathize easily with the end-user and are good at capturing and expressing emotions. They create new things and are used to creating mock-ups and sketches to visualize their ideas and share them with others, which is key for experimenting and testing.
- Business people — research existing market offerings. They perform competitive analysis, collect data about the size of the market and interview experts looking for trends, either social, environmental, technological or political, and read industry reports. They provide a recap of findings in presentations, with market forecasts and make strategic recommendations, supported by calculations on expected ROIs.
- Engineers/Scientists — conduct technological research, researching patents, reading recent findings, technical documents and potential sources for product/service development. They analyze crash reports and conduct calculations. They sum up findings with technical notes and reports and suggest alternative technologies or designs.

With the descriptions above, on the one hand, designers look for subjectivity, human experience and emotion in their work. Their approach is qualitative, experiential, intuitive, and novel.

The intuitive mind is a sacred gift and the rational mind is a faithful servant — Albert Einstein

But novelty does not necessarily create value. And even if a value is theoretically created, it needs to be executed to generate profits for the business. Therefore, business people, engineers, and scientists are used to think rationally and objectively based on quantitative data. Their objective is to reduce uncertainty by pursuing control and stability.

Steve Jobs on Failure

In design thinking, one core strength is that it's okay to be wrong, to be breaking things, it's okay to put something out there and have people critique and challenge it, because there's a nugget inside something failing, and design thinkers will only find the 'right — solution' after failing.

Design thinking is 'intuitive — thinking'

Design thinking is a systematic approach that enforces the combination of intuitive (designers) and analytical thinking (business people and engineers).

On the pattern of reasoning in innovative design

Simulating the design process of artifacts by computers requires the modeling of design reasoning in a logical...

When it comes to analytical thinking, there are two lines of reasoning in:

- **Deductive reasoning** — is useful to make sense of new information, based on existing knowledge. ‘Deduction’ is the logical reasoning that, using a general rule, infers a conclusion after observing an event or finding evidence. The crucial task with deductive reasoning is to select the appropriate rule to apply and then confirm or refute the conclusion.
- **Inductive reasoning** — is logical reasoning that infers a rule-based upon observations of a cause and effect. The inferred rule is credible given the premises and follows the theory of evidence. In inductive logic, any new evidence consistent with existing beliefs and mental models reinforces the beliefs and mental models, while other evidence is simply considered anomalous. Therefore, inductive logic can prevent individuals from interpreting evidence in alternate ways; it lacks intuitive thinking.

But design thinking starts with intuitive thinking — and intuitive thinking is based on abduction, reasoning that infers and proves an explanatory hypothesis for a body of recently discovered observations. For example, abduction happens when a product manager suggests an explanatory hypothesis to account for the patterns and insights observed in the user behavior data collected on a mobile application. If the hypothesis is validated, it becomes a rule that feeds deductive thinking for a larger group of users.

Deduction, Induction, Abduction by GeorgiaTech

There are two types of abductive reasoning:

- **Explanatory abduction** — is used when we observe an anomalous or surprising event, by which we infer a hypothesis to explain the new observation. Explanatory abductions give sense to new observed behavior and data collected.
- **Innovative abduction** — is used when we don’t know the means to achieve the desired outcome. Here we infer two hypotheses, one that connects the strategic option to the desired outcome and one that connects the strategic option to its implementation or mode of operation. The desired outcome is the only given or known data point. The strategic option explains what should exist to achieve the desired outcome and the mode of operation explains how the strategic option will be realized and executed. Innovative abductions lead to a strategic option and its implementation method to deliver a new outcome.

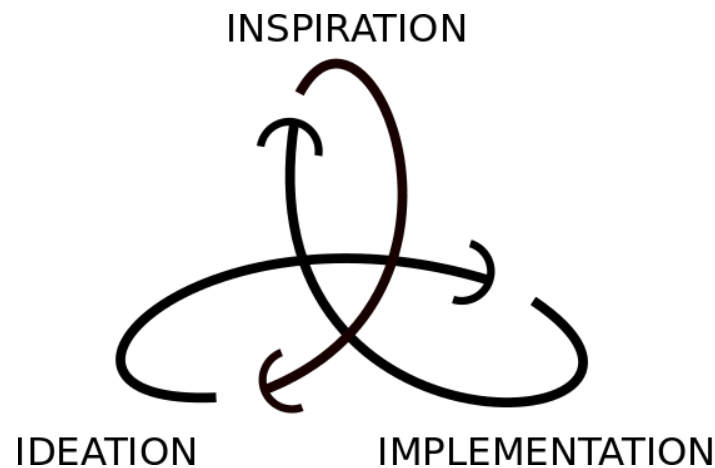
For example, consider we want to improve the citizens’ recycling behavior. Here, we first start by developing empathy with the citizens and observing their recycling behaviors. This

observation reveals a potential explanatory hypothesis or insight: ‘People will only recycle if the product is useful to someone.’ At this stage, the formed insight is neither true nor false, it’s a hypothesis that needs to be tested. For that purpose, you run experiments to generate data on the motivations for recycling, experiments on recycling with and without intrinsic and extrinsic incentives. As a result, the hypothesis is validated, allowing you to trust your hypothesis and further extend it to all other citizens. This is the power of abductive reasoning.

DESIGN THINKING IN ACTION

The 3-phases of design thinking

There are three phases to design thinking:



IDEO's 3 phases of design thinking

- **Inspiration** — this involves going in the field, interviewing several segments of stakeholders, observing them, living their experience, collecting data with pictures, videos, interviews, and taking it back to discuss within the team until patterns and hypotheses emerge. These findings should explain the users' behavior and help them understand their needs and pains.
- **Ideation** — this refers to the generation of ideas or solutions to address user problems and needs. These ideas are combined in comprehensive solutions that could potentially address the needs and pains.
- **Implementation** — in this phase, teams rapidly build rough and low-cost prototypes to put concrete solutions into the hands of users, thereby testing problem-solution-fit.

Duration of phases

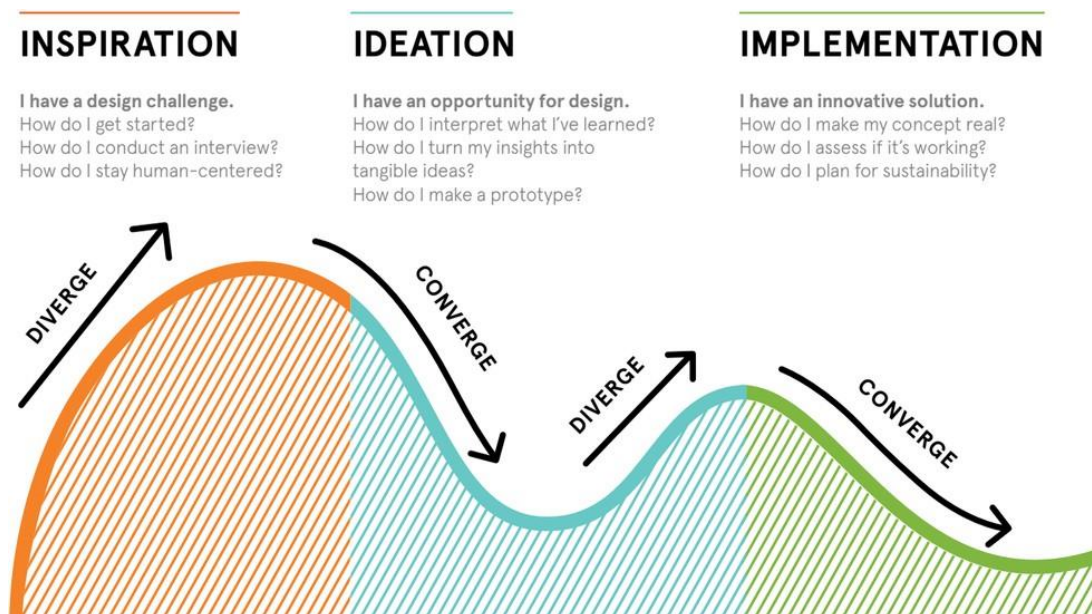
Inspiration and implementation phases take the longest to complete:

- The inspiration phase is concrete, it's about understanding the users and their pains and problems and takes time as teams need to deep dive into users' mindset and behaviors.
- Ideation is abstract and imaginative. It needs focus and it's intense. But it should not take a large amount of the team's time. The more time is spent here, the less we can interact with users to test the solutions.
- Implementation is once again concrete. It's about experimenting by confronting ideas with reality. Teams put the prototypes in the hands of the users. Here again, it takes time to build the prototypes, test them, gather feedback, and reiterate designs.

The ideation funnel

During the design thinking process, it is common to generate as many as 250 ideas, which are then narrowed down to as few as 25 concepts. This widening and narrowing at each phase are called divergent and convergent thinking.

- **Divergent thinking** is about opening the choices, looking broadly and considering many different options. The rationale of divergent thinking is to avoid being trapped in the usual setting of the problem within a pre-existing set of solutions.
- **Convergent thinking** is the process of narrowing down options and selecting the most promising ones to be further explored and tested.



IDEO Human-Centered Design Process — Image by source: Eventbrite.com

The people and the environment

Design thinking's guided and systematic process is followed by people from different backgrounds, skills, and expertise, gathering in a specific place that favors sharing and building a constructive and collective mindset. In other words, the design thinking process is sequenced and clear, but it has to be contextualized to the environment of the organization it is immersed in. Within the design thinking process and in the long term, people and the environment they are working in is the most important pillar to successful design thinking efforts. After all, one needs talented and happy designers to run design thinking activities.

Visualization is key to success

Visualization is the transformation of information into images. It makes data and ideas tangible, accessible and interpretable with emotion and intuition. Designers favor drawing over text because while the text is open to various interpretations according to your interlocutor, a drawing reduces the possibility of mental models. This is why startups build user personas in their early days when trying to define their customers.

Visualization is important at every phase of the design thinking process:

In the inspiration phase, photographs and sketches are used to capture the observations. Visual thinking is also required to synthesize and structure data and look for patterns and insights. In the ideation phase, the brain is creating images, and concepts are coming out of our imagination. In the implementation phase, more detailed diagrams are needed for prototype development and to represent how the user will interact with the solution to identify the main points of vigilance and friction.

Storytelling

Just like a comic, a story has a narrative that involves emotions, enabling people to understand issues and create engagement. Storytelling is the human way of conveying information. And since design thinking is a human-centered approach, every phase of the process needs storytelling.

A story adds context to the data gathered. The inspiration phase is about listening to and interpreting human stories, writing them down, and developing empathy with user personas and pains. In the ideation phase, storytelling makes ideas real, concrete, and tangible. In the implementation phase, storytelling on-boards stakeholders and engages them to have their feedback and comments included in the suggested solution.

How to Create Stories (and Products) People Want — with Shonda Rhimes and Marc Andreessen

Project types best suitable for design thinking

Design thinking is not adapted for all types of innovation projects but can work wonders in any project that needs to connect deeply with the customers, transform insights and data into actionable ideas, or create and implement new solutions with fast and effective business impact.

There are five criteria to consider when deciding to use design thinking:

- The project needs to be human-driven
- The project has a high degree of uncertainty. Uncertainty is a situation where its probability of occurrence and degree of impact are unknown.
- The project has a poorly defined problem
- The project is complex, meaning it has interdependent and dynamically moving parts and therefore can't be modeled using links between causes and effect. So it is difficult to know where to start.
- It lacks data or the available data is not organized and well-framed

Therefore, if humans are not at the heart of the problem, if there is a clear and shared definition of the problem if you can accurately predict and determine cause and effect in advance, if the problem is decomposable into parts, and if you have well-framed and structured data, then design thinking is not suitable for such a project.

Design thinking is an approach that we can follow to reduce uncertainty levels through experimentation and testing and not through prediction — which is generally used in risk management projects. Design thinking is suitable and should be adopted when there are many stakeholders involved and everyone has their definition of the problem. When time and effort are required to learn about the context of the problem. When it is not possible to prove that any particular solution will work until it is tested and when several expertise and specializations are required over a collaborative team effort.

EXPLORE PRESENTATION SIGNERS ACROSS GLOBE

MVP OR PROTOTYPING

Prototypes and Minimum Viable Products (MVP) are a critical part of product development. Sure, you could dive right in, spend months developing an expensive product with all the bells and whistles. Only to find that at launch, nobody wants it. Or you've built it completely wrong. Or you could have built something way better if you'd pivoted halfway through.

Not testing your ideas is a great way to fail fast. That's where MVPs and prototypes come in! They help you to find your product-market fit, get stakeholder and user feedback, and start building buzz around your upcoming launch. They're also a great communication tool between you and your designers/tech teams, and they help you understand what it is you're building. But what is the meaning of 'prototype' and 'MVP'?

Well, an MVP is a lightweight early-version of your product. A prototype is also a lightweight early-version of your product.

Done! ...if only it were that simple. While in theory MVPs and prototypes seem like the same thing, and both are part of agile product development, there are a few key differences.

Here we're going to take you through everything you need to know about prototypes and MVPs. We'll look at:

- What a prototype is for, and the forms they can take.
- What an MVP is.
- Examples of MVPs that became huge, successful products.
- The main differences between a prototype and an MVP.
- Tips on how to build your own.

What is a Prototype?

A prototype is the theory behind your product made into form. Instead of having a vague idea of what you want to eventually build, you make something tangible.

Prototypes can vary in size and how much time and effort you put into them. It might be a simple sketch on paper or it might be something more functional and interactive.

The main benefit of a prototype is that it helps you to communicate with your design and engineering teams. You need to show them what to build! Think of it as the packaging of an empty box. It shows you what's inside and what it looks like, but there's currently nothing going on underneath.

Depending on your skill level, you can make your prototype as complicated or as simple as you like.

Different Types of Prototypes

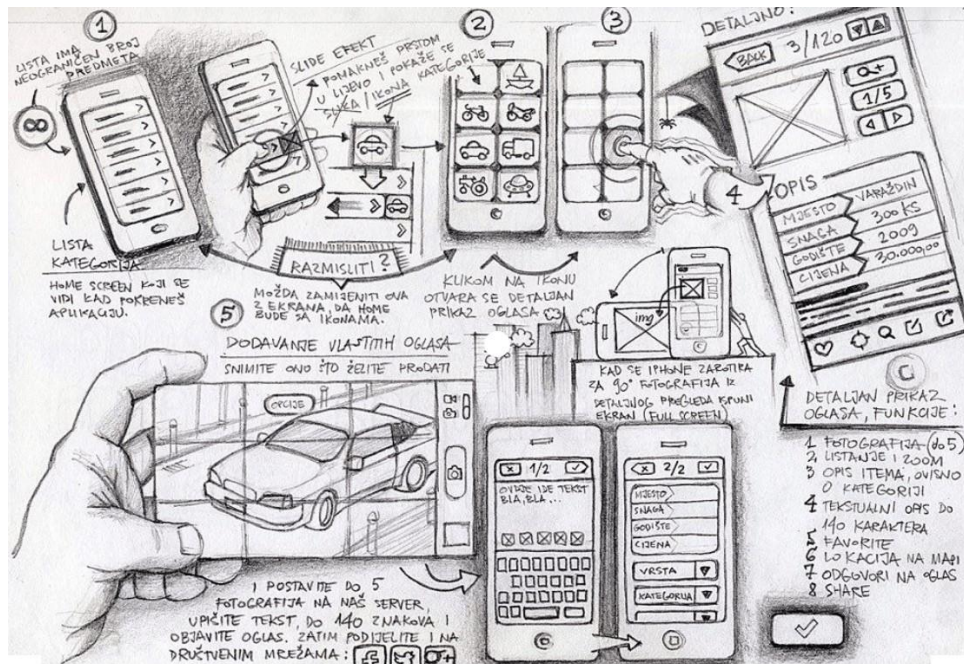


Image credit: Sunny Cui, 'A Guide to Paper-Prototype Sketching'

Paper sketches: This is the lowest level of prototype. If you can wield a pencil, you can manage! Draw out a basic idea of what your user interface will look like. If you want to make something visually appealing to communicate to investors/stakeholders, work with a designer to create something that fits your vision.

Digital prototypes: You can also create a working mock-up of your envisioned product, and there are many tools to help you do so.

The rule of thumb when picking prototyping tools is their complexity: you might not need all the features in the world to make your project scream SUCCESS.

For a web-based app, if you're just starting out and don't have a big budget/powerful computer, we can recommend [Proto.io](#). In terms of standalone apps, [InVision](#), [Figma](#), [Balsamiq](#), [Axure RP](#), [Sketch](#) (for Mac only) and [Adobe XD](#) are good options.

Prototype Inspiration

Need some ideas? Check out these awesome prototype examples!

- Household Energy Monitor
- Framer Web Beta
- Stay Home App

What is an MVP?

The meaning of MVP is Minimum Viable Product. It's the bare minimum version of your product that still solves the problem for your customers. Eric Ries describes it perfectly in *The Lean Startup*.



MVP is the version of a new product that allows a team to collect the maximum amount of validated learning about customers with the least effort

Eric Ries, *The Lean Startup*



Think of it like this: if someone needed transportation between Point A and Point B, and they're not sure how best to get there, the first thing you give them wouldn't be a Tesla! You might start with a skateboard, or a scooter. You get the feedback that they want something motorized, so you get them a motorbike. It does the job, but now they want something waterproof, with some kind of entertainment system. So eventually you give them a car with built-in radio.

Without testing, and understanding what works and what doesn't, you might have given them a skateboard with a Nintendo Wii strapped to the front instead.

Examples of MVPs that became products

1. Groupon: A WordPress site and PDFs

While they didn't exactly invent the voucher (extreme couponing has been a hobby for generations!) Groupon helped it become a staple of our social lives and shopping habits. It started out as a real bare-bones operation. At launch, they had a very basic WordPress site, and would email normal PDFs as vouchers to subscribers. Once they proved that there was a demand for an online voucher marketplace, they invested in building a more elaborate backend. As of 2018, they have sold 1.5 billion vouchers to almost 50 million annual active users. To put it lightly...not bad.

2. Airbnb: From 3 guests to 400 million

The Airbnb we know and love today, started out as AirBed&Breakfast, a very basic website which provided accommodation to people who came to San Francisco for a design conference.

The problem that founders Brian Chesky and Joe Gebbia were trying to solve was that they couldn't afford the rent on their loft apartment. Short-term rentals were arduous and often involved big bills to middlemen.

After testing their idea on 3 paying guests during the conference, Airbnb grew organically to the powerhouse we know today. It even involved Brian Chesky staying in early listings to give the platform some traction.

Had they not tested, they could have built an extravagant system, with all the cool features Airbnb has in 2020, only to find that people didn't want to stay in other people's homes. Luckily for them, they were onto something...

3. Dropbox: The product that didn't exist

Dropbox is a fantastic example of product-led growth done well. But in the beginning, they didn't have a product at all!

Instead of building a full product, they came up with a working MVP, created a video explainer, and essentially pretended that it existed.

This helped them to test their idea. Had they built the complicated backend, only to find that nobody wanted their product, they would have lost countless hours of work.

From their introductory video, they received over 70k email addresses from people wanting Dropbox ASAP. They'd proved their idea had legs without spending a lot of money getting it to walk.

4. Spotify: Do one thing and do it well

When Spotify first launched its MVP to the world, it aimed to do one thing: stream music. They built an MVP which only streamed songs, without all of the extra features (playlists, podcasts, video, mobile app etc), and launched a closed beta to test their idea. Everyone loved it, and the freemium model was popular with users.

Knowing that their MVP was a hit, and their business model was proven, they rolled out a full launch and have experienced year on year growth ever since.

At the end of Q1 in 2020, Spotify had 130 million premium subscribers.

What is The Difference Between Prototype and MVP?

If we had to break it down very simply, we could say that the main differences are:

1. **Scope**
2. **Commitment**
3. **Audience**

The scope of a prototype and an MVP vary greatly. A prototype is something that you can put very little time and effort into. The idea is to come up with a variety of prototypes, and be able to change, adapt, and throw away ideas very quickly.

With an MVP, you've picked an idea and you're rolling with it, which means putting in a little more effort. You're likely to use more resources and have more people involved.

Because of the difference in scope, your commitment level is also different. Once you've invested time and resources into an MVP, you're less likely to completely scrap it. While the whole concept of MVP is based on the ability to be cost-effective and make rapid changes, it'll still be painful to get rid of it entirely and start from scratch.

With a prototype, it can be as easy as crumpling up the paper and starting again! Although more complex prototypes certainly take up more resources, you're still less tied to them than you are to a fully realized MVP.

Your prototypes and your MVPs will also be intended for different audiences. Prototypes are mostly internal. Mainly intended for your teams, or others in your company. At most, you'll give them to the press to create a bit of buzz, or use them for crowdfunding purposes. But they're not intended for your target market en masse.

An MVP, as we've now seen, is intended to be launched to your target market in order to assess its reception. The P stands for Product after all. An MVP is something that the public will be judging you on, and something which people may remember when the time comes to commit to V2.

Tips for Building Your Prototype or MVP

Now you know the difference, you might be thinking about how to build a prototype or MVP. While the resources and data you'll need are different for each, there are a few golden rules that you can apply, no matter what you're building.

1. Stay customer-focused: So many of the world's most successful companies credit their customer-focus as the secret to building winning products. The goal of your product is to provide value, so be sure to focus on who you're building for.

2. Stay data-driven: You don't have to be a data scientist to be a product manager, but don't think that numbers shouldn't matter to you. You might not have an extensive data set if you're building your first ever product, so think about what data you want to collect from your prototype/MVP. It'll help you later in development, and for future releases.

3. Kill your darlings: You might really love your prototype/MVP. But if no one else does...it's time to rethink! The only thing waiting for you if you launch something nobody wants, is failure.

4. Just build the thing! You can agonize over perfecting your prototype before committing to an MVP. You can also spend months working and reworking your MVP before you feel ready to launch it. If you want to build products, eventually it's time to let go and let people see them!